

Deliverable 1

What is a web server?

A software that looks for incoming requests from clients (like web browsers) via protocols such as HTTP or HTTPS and then delivers the requested content.

What are some different web server applications?

Some different web server applications are:

- Apache HTTP Server - A widely used, open-source web server known for its stability, modularity, and support for various operating systems.
- Nginx - A high performance, event-driven web server known for handling high traffic loads efficiently, often used as a reverse proxy and load balancer.
- LiteSpeed - A web server designed as a drop in replacement for Apache
- Caddy - A web server focused on automatic HTTPS configuration and simplicity
- Microsoft IIS - Designed specifically for Windows server environments
- Apache Tomcat - An open-source Java Servlet container and web server used mainly to run Java-based applications and servlets
- Lighttpd - A fast web server optimized for low memory usage and high speed, suitable for servers with limited resources
- Node.js - A JavaScript runtime environment built on Chrome's V8 engine that can act as a web server
- Jetty - an open-source Java-based web server and servlet container, often used for embedding in applications
- Hiawatha - A secure web server for Unix systems, known for its simplicity and built-in security features like anti-flooding and CGI handling

What is virtualization?

A technology that creates virtual versions of computing resources, such as servers, storage, networks, or entire computers.

What is virtualbox?

A free, open-source, Type 2 hypervisor developed by Oracle Corporation that enables users to run multiple operating systems simultaneously on a single physical machine.

What is a virtual machine?

Something that emulates a physical computer, running its own operating system and apps with virtualized resources. It's isolated from the host system, allowing users to perform secure tasks like testing apps or using different operating systems while optimizing physical hardware.

In the context of virtualization, what does host machine and guest machine mean?

The **host machine** is the physical hardware that runs the virtualization software and provides essential computing resources such as CPU, memory, storage, network I/O to virtual environments. The **guest machine** is the virtual machine created on the host, which operates as an independent instance with its own operating system and applications but relies entirely on the host for hardware resources.

What is Debian?

A free, open-source Linux distribution developed by the Debian Project, a global community of volunteers founded by Ian Murdock in August 1993.

What is a firewall?

A network security device or software application that monitors and controls incoming and outgoing network traffic based on a predetermined set of security rules.

What is SSH?

A cryptographic network protocol designed to operate network services securely over an unsecured network.

What is an IP Address?

A unique numerical label assigned to every device connected to a computer network that uses the Internet Protocol for communication.

What is a network mask?

A 32-bit value used in IPv4 networking to divide an IP address into a network portion and a host portion.

What is a port? (In the context of networking/computers)

A virtual endpoint that identifies a specific process, service, or application, running on a device, allowing the operating system to direct incoming and outgoing data traffic to the correct program.

What is port forwarding?

A networking technique that allows external devices to access services or devices on a private local area network by redirecting incoming traffic from a specific public port to a designated internal IP address and port.

What is localhost? (In the context of networking/computers)

A hostname that refers to the local machine currently making a request, allowing it to communicate with its own network services without using an external network or internet connection.

What does this ip address represent 127.0.0.1?

The standard IPv4 loopback address, commonly referred to as localhost. It represent the computer itself, allowing the device to communicate with its own network services without sending data out to an external network or physical network interface card.

What is Git?

A free, open-source, distributed version control system created by Linus Torvalds in 2005 for the development of the Linux kernel.

What is GitHub?

A cloud-based, web-based platform own by Microsoft that allows developers to store, manage, track, and collaborate on code.